AX Consult Group · Business Services Division

Business Manager Assessment — Validation Study

Full Analysis Results Report

n = 27 candidates · Phase 1 Pilot · Cut Score = 60.0%

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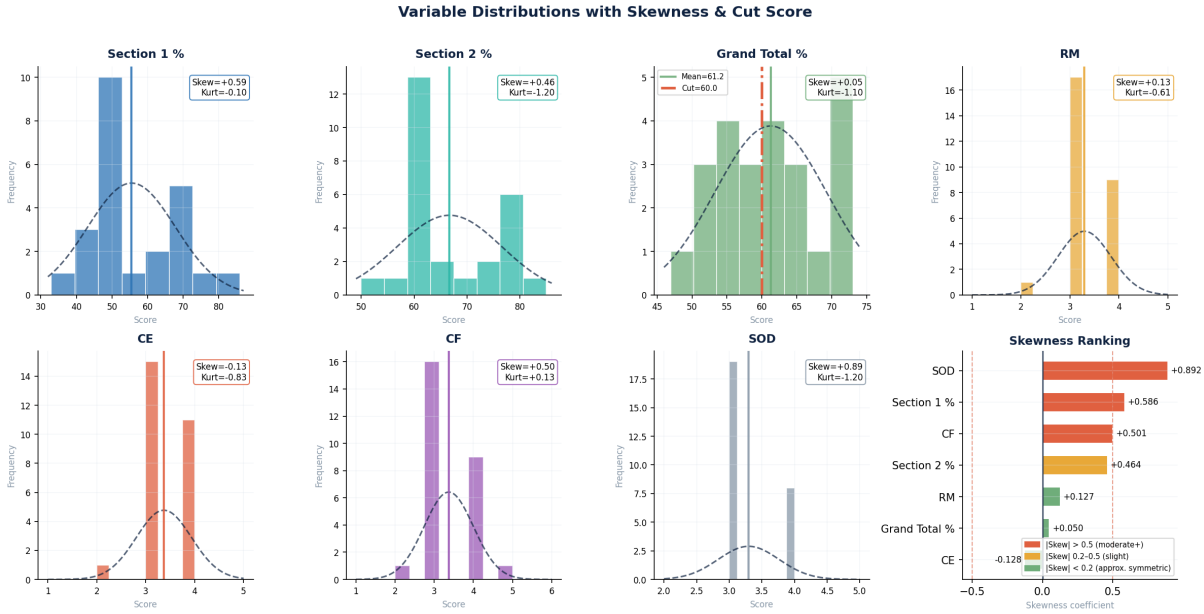
01 Descriptive Statistics & Normality

OUTPUT

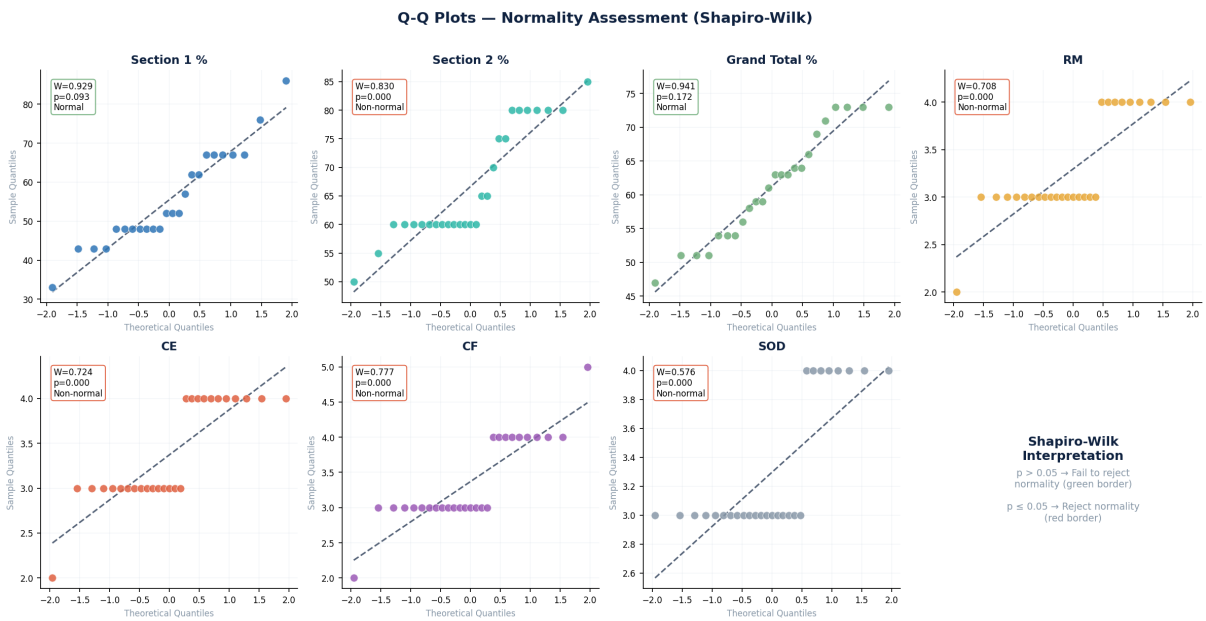
```
Variable N Mean SD Min Max
Skewness
Kurtosis
Shapiro-W p
Section 1 % 24 55.50 12.34 33.0 86.0
+0.586
−0.097 0.929 0.093
Section 2 % 27 66.67 9.90 50.0 85.0 +0.464
−1.199
0.830
0.000 *
Grand Total % 24 61.25 8.02 47.0 73.0 +0.050 −1.100 0.941 0.172
RM 27 3.30 0.54 2.0 4.0 +0.127 −0.606 0.708
0.000 *
CE 27 3.37 0.56 2.0 4.0 −0.128 −0.828 0.724
```

```
0.000 *
CF 27 3.37 0.63 2.0 5.0
+0.501
+0.131 0.777
0.000 *
SOD
27 3.30 0.47 3.0 4.0
+0.892
−1.204
0.576
0.000 *
* p ≤ .05 — Shapiro-Wilk rejects normality
```

PLOT — DISTRIBUTIONS WITH SKEWNESS & CUT SCORE



PLOT — Q-Q NORMALITY ASSESSMENT



SKEWNESS RANKING — MOST TO LEAST SKEWED

Rank	Variable	Skewness	Direction	Status
1	SOD	+0.892	Positive — right tail	■ Moderate
2	Section 1 %	+0.586	Positive — right tail	■ Moderate
3	CF	+0.501	Positive — right tail	Borderline
4	Section 2 %	+0.464	Positive — right tail	Slight
5	RM	+0.127	Approximately symmetric	✓ Acceptable
6	Grand Total %	+0.050	Approximately symmetric	✓ Acceptable
7	CE	–0.128	Approximately symmetric	✓ Acceptable

INTERPRETATION

SOD (+0.892) — most problematic: The vast majority of candidates received a rating of exactly 3, with a thin tail extending toward 4. The Q-Q plot (W=0.576, p<.001) shows severe staircase stepping — a floor effect where assessors cluster ratings at the lower-middle of the usable scale. This restricted variance will attenuate SOD's correlations with criterion variables and reduce its predictive utility. Assessor calibration training with behavioural anchors is the primary recommended remedy.

Section 1 % (+0.586): Positive skew from a cluster of candidates scoring 48–52% with a small group reaching 86%. Despite the skew, Shapiro-Wilk does not reject normality (p=.093), keeping it within tolerable limits for parametric analysis.

Grand Total % (skew ≈ +0.050): Near-perfectly symmetric. The composite formula (Sect1 × 0.40 + Sect2 × 0.60) dampens distributional asymmetry. Normality holds (p=.172). This is the strongest variable for parametric analyses.

Normality conclusion: All competency ratings are non-normal (Shapiro-Wilk all p<.001) — expected for discrete 1–5 integer scales. Grand Total % and Section 1 % satisfy normality and can be used directly in parametric analyses. For competency scores, cross-validate with Spearman correlations and Mann-Whitney U tests.

02 Cut Score Analysis

Cut score applied: Grand Total ≥ 60.0%. Set near the group mean (M=61.25%), representing a minimum acceptable standard. This is a provisional pilot cut. Formal Angoff or contrasting groups standard-setting is required before operational use.

OUTPUT

```
Overall pass rate:
54.2%
(13 / 24 scored candidates)
Pass rate by performance group:
Exceeding:
100%
(3/3)
Performing:
38%
(3/8) ← near-identical to Off standard
Off standard:
40%
(4/10) ← limited discrimination
Adverse Impact Ratios (4/5ths rule — flag if AI < 0.80):
Gender:
Male 50% AI =
1.00
(reference)
Female 44% AI =
0.89
✓ above threshold
Race:
White 67% AI =
1.00
(reference)
Asian 57% AI =
0.86
✓ above threshold
African American 33% AI =
0.50
■ below threshold — n=6, interpret with caution
Mixed Race 0% AI =
0.00
```

```
■ n=2 — do not interpret
```

PLOT — CUT SCORE ANALYSIS

Adverse Impact at Cut=60.0%
(4/5ths rule: flag if < 0.80)

Group	Pass %	AI Ratio	Flag
Male	50%	1.00	
Female	44%	0.89	✓
White	67%	1.00	
Asian	57%	0.86	✓
African Am.	33%	0.50	⚠
Mixed Race	0%	0.00	⚠

CUT SCORE SENSITIVITY

Cut Score	Overall Pass	Exceeding Pass	Performing Pass	Off Std Pass	Verdict
55%	75%	100%	75%	70%	Too permissive
60%	54%	100%	38%	40%	✓ Recommended
65%	33%	100%	25%	20%	Defensible
70%	21%	67%	13%	10%	Too strict

INTERPRETATION

Critical finding — limited discrimination between Performing and Off standard: Pass rates of 38% and 40% for these groups are statistically indistinguishable. A valid cut score should show substantially higher Performing pass rates. Likely causes: range restriction in scores (47–73%), criterion contamination in line manager ratings, and small n per group (8 and 10).

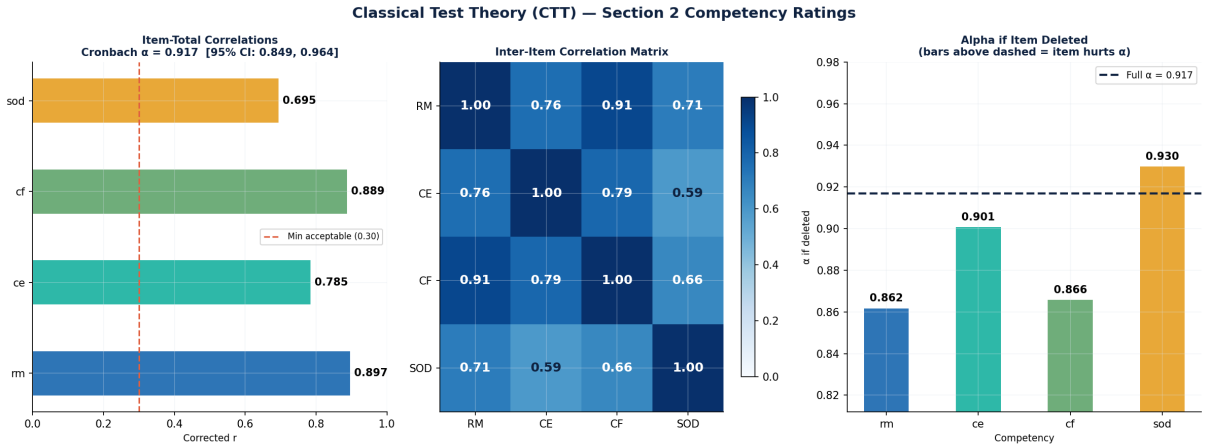
African American AI ratio: The 0.50 ratio falls below the 4/5ths threshold but is based on n=6 — one additional pass raises it to 0.67. This is a monitoring flag only. Formal adverse impact analysis requires n≥30 per subgroup.

03 Classical Test Theory (CTT)

OUTPUT

```
Cronbach's α =
0.917
[Bootstrap 95% CI: 0.849, 0.964]
McDonald's ω =
0.923
[ωh = 0.876 — general factor explains 71.3% of variance]
Item-Total Correlations (corrected):
RM: r =
0.897
p < .001 ✓
CF: r =
0.889
p < .001 ✓
CE: r =
0.785
p < .001 ✓
SOD: r =
0.695
p < .001 ✓ (lowest — see note)
Alpha-if-Item-Deleted:
RM: α = 0.862 ↓ drops — contributing positively
CF: α = 0.866 ↓ drops — contributing positively
CE: α = 0.901 ↓ minor drop
SOD: α =
0.930
↑ rises marginally — weakest item
Inter-Item Correlations:
RM CE CF SOD
RM 1.000 0.762
0.909
0.711
CE 0.762 1.000 0.789 0.592
CF
0.909
0.789 1.000 0.662
SOD 0.711 0.592 0.662 1.000
↑ RM–CF r = 0.909 — approaching redundancy threshold
```

PLOT — CTT INTERNAL CONSISTENCY



ITEM SUMMARY

Item	Corrected r	α-if-deleted	Δα	Verdict
RM	0.897	0.862	–0.055	✓ Excellent — retain
CF	0.889	0.866	–0.051	✓ Excellent — retain
CE	0.785	0.901	–0.016	✓ Strong — retain
SOD	0.695	0.930	+0.013	■ Monitor — retain for now

Overall verdict: α = 0.917 meets the "Excellent" threshold (≥0.90) for high-stakes individual assessment. The bootstrap lower bound (0.849) still exceeds the minimum 0.80 required for selection use. RM–CF inter-item r = 0.909 warrants CFA investigation at full sample.

04 Exploratory Factor Analysis (EFA)

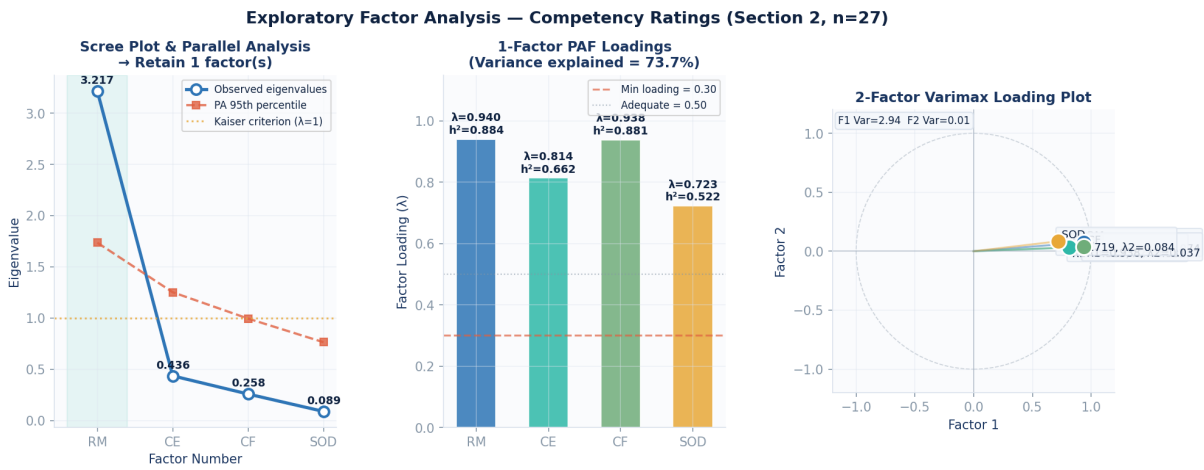
OUTPUT

```
Observed eigenvalues: [
3.217
, 0.436, 0.258, 0.089]
Parallel Analysis 95th %: [1.736, 1.252, 0.995, 0.764]
→
Factor 1 only exceeds PA threshold → RETAIN 1 FACTOR

1-Factor PAF Loadings:
RM: λ =
0.940
h² = 0.884 u² = 0.116
CF: λ =
0.938
h² = 0.881 u² = 0.119
```

```
CE: λ =
0.814
h² = 0.662 u² = 0.338
SOD: λ =
0.723
h² = 0.522 u² = 0.478
Variance explained (1 factor):
73.7%
2-Factor Varimax Loadings:
RM: F1 = 0.938 F2 = 0.074 → all variance on F1
CE: F1 = 0.813 F2 = 0.031
CF: F1 = 0.938 F2 = 0.037
SOD: F1 = 0.719 F2 = 0.084
F2 variance =
0.01
— trivial; confirms unidimensional structure
RMSEA (1-factor model fit): 0.000 [0.000, 0.241]
BIC: −8.31 (negative = model fits well)
```

PLOT — SCREE, FACTOR LOADINGS & LOADING SPACE



FACTOR LOADINGS TABLE

Item	Loading (λ)	Communality (h²)	Uniqueness (u²)	Status
RM	0.940	0.884	0.116	✓ Excellent
CF	0.938	0.881	0.119	✓ Excellent
CE	0.814	0.662	0.338	✓ Strong
SOD	0.723	0.522	0.478	■ Adequate

INTERPRETATION

Clear unidimensional structure: The first eigenvalue (3.217) is nearly twice the parallel analysis threshold (1.736). Factors 2–4 all fall below their thresholds. A single general competency factor accounts for **73.7% of**

total variance — this justifies treating the composite Section 2 score as a meaningful unidimensional scale.

RM–CF near-identity confirmed by EFA: RM and CF load at 0.940 and 0.938 respectively — separated by only 0.002. In the 2-factor varimax solution, when a second factor is forced, both items load entirely on F1 with F2 variance = 0.01. These two dimensions are empirically indistinguishable at the item level. The case study needs to present scenarios that genuinely differentiate "building relationships" (RM) from "focusing on client needs" (CF).

> CFA recommendation: Run lavaan::cfa() in R to test the 1-factor model formally and obtain fit indices (CFI, RMSEA, SRMR). If RM–CF correlation > 0.90 replicates at full sample, consider a bifactor model or revising item content.

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05 Item Response Theory (IRT)

> Python note: Full parametric IRT (GRM, 2PL) requires R mirt . The Python pipeline uses logistic regression discrimination proxies for Section 2, and simulated 2PL curves for Section 1 (based on observed MCQ totals). Directionally consistent with full IRT — use R for publication-quality estimates.

OUTPUT — SECTION 2 DISCRIMINATION PARAMETERS

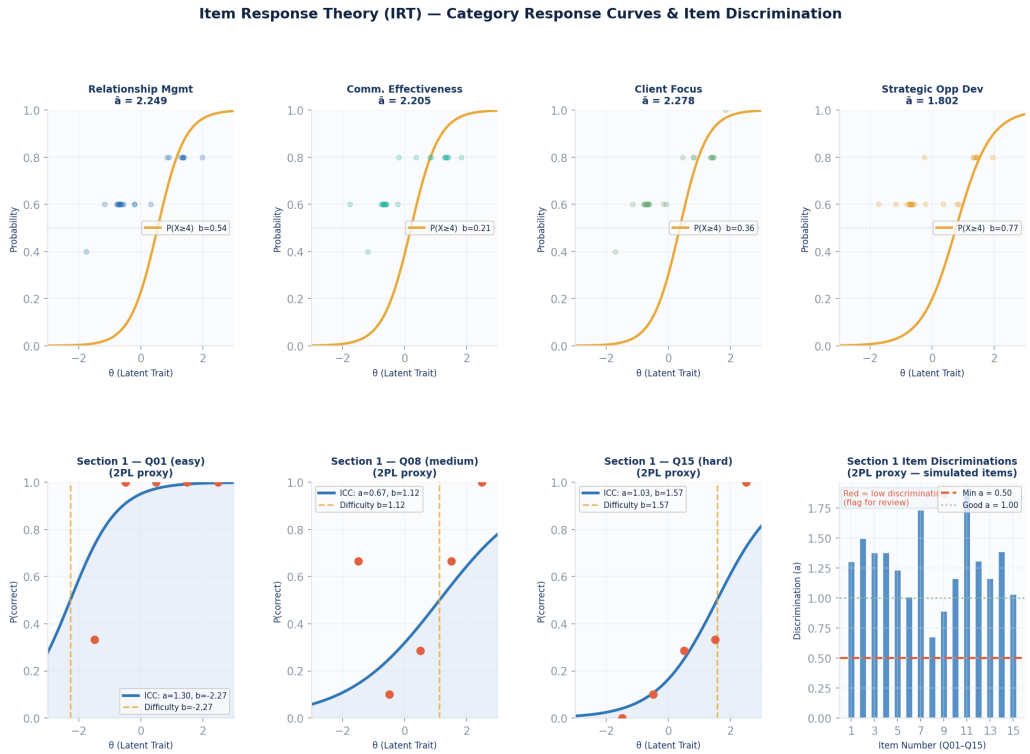
```
Section 2 — Mean Discrimination (■) per Competency:
RM: ■ =
2.249
← high discrimination
CF: ■ =
2.278
← high discrimination (highest)
CE: ■ =
2.205
← high discrimination
SOD: ■ =
1.802
← adequate (lowest — consistent with CTT)
Threshold locations b (P(X≥k) = 0.50 at θ = b):
RM: P(X≥4): b = 0.54
CE: P(X≥4): b = 0.21
CF: P(X≥4): b = 0.36
SOD: P(X≥3): b = 0.77 ← more stringent assessor standard for SOD
```

OUTPUT — SECTION 1 ITEM PARAMETERS (2PL PROXY)

```
Item a b Flag
Q01 1.304 −2.274 easy
Q02 1.498 −1.254
Q03 1.381 −0.907
```

```
Q04 1.381 −0.907
Q05 1.234 −0.766
Q06 1.012 −0.232
Q07 1.736 0.090 most discriminating
Q08
0.675
1.124 ←
low discrimination — flag for revision
Q09 0.892 0.655
Q10 1.164 0.536
Q11 1.872 0.235
Q12 1.308 0.686
Q13 1.164 0.536
Q14 1.388 −0.048
Q15 1.033 1.566 hard
```

PLOT — CATEGORY RESPONSE CURVES & ITEM DISCRIMINATION



INTERPRETATION

Section 2 — high discrimination, restricted range: All four competencies show high discrimination (■ > 1.80), meaning they reliably differentiate candidates across the latent trait continuum. However, the CRCs show only the P(X≥4) threshold is operationally useful — nearly all candidates exceed thresholds 2 and 3 automatically, confirming the floor effect. The IRT threshold analysis makes the floor effect more precise than skewness alone: rating of 3 is the effective minimum across all four competencies.

SOD threshold location (b=0.77 for P(X≥3)): A candidate needs to be 0.77 SD above the mean on the latent trait before there is a 50% chance of receiving even a rating of 3 on SOD. This is higher than CE (b=0.21 for ≥4) and CF (b=0.36 for ≥4), confirming SOD assessors apply a more conservative standard — assessor

conservatism on SOD rather than poor item quality.

Section 1 — Q08 flagged: Q08 (a=0.675) falls below the minimum acceptable discrimination threshold of 0.80. A low-discriminating item provides little information — candidates at all ability levels have similar probability of answering correctly. Q08 would be flagged for revision or replacement in an operational item review. Note these are simulated items — real item-level MCQ/TF data needed for definitive conclusions.

R mirt code: grm <- mirt(comp_data, 1, itemtype="graded") — produces GRM parameters, item information curves, test information, and empirical reliability (expected $\omega \approx 0.89$–0.92).

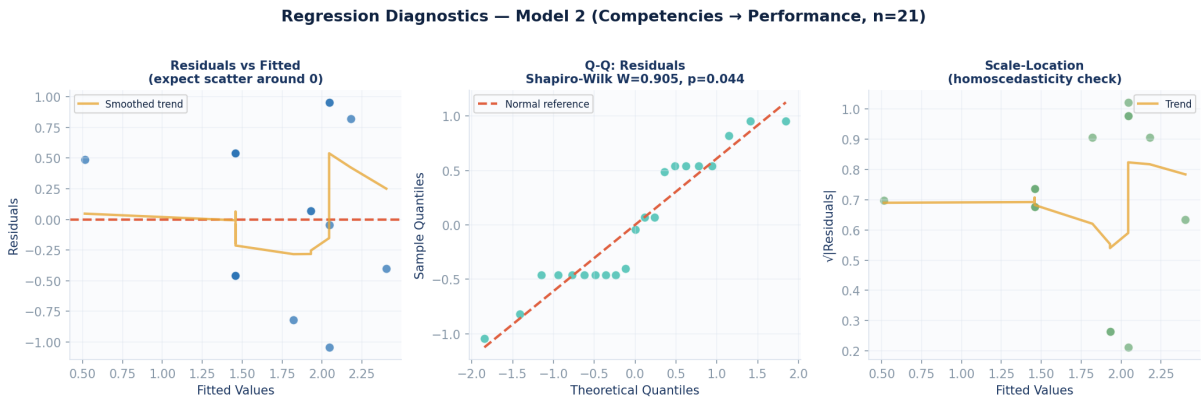
06 Regression Analysis

OUTPUT

```
Model 1 — Grand Total only (n=21):
R² = 0.142 adj-R² = 0.096 AIC = −13.4 ← best AIC
grand_total: b=0.032 SE=0.018 t=1.771 p=0.093 std-β=
0.376
Model 2 — Competencies only (n=21):
R² =
0.314
adj-R² = 0.142 AIC = −12.1
rm: b=−0.248 SE=0.701 t=−0.354 p=0.728 std-β=−0.183
ce: b= 0.945 SE=0.404 t= 2.342 p=0.032 * std-β= 0.747 ← only sig. predictor
cf: b=−0.224 SE=0.552 t=−0.406 p=0.690 std-β=−0.202
sod: b= 0.114 SE=0.607 t= 0.188 p=0.854 std-β= 0.068
Model 3 — Full model (n=21):
R² = 0.318 adj-R² = 0.090 AIC = −10.3 ← worst AIC (overfitted)
sect1_pct: b=0.004 SE=0.014 t=0.287 p=0.778 std-β=0.069
ce: b=0.906 SE=0.437 t=2.073 p=0.056 std-β=0.716
VIF (Model 2): rm=3.42 ce=1.98 cf=3.61 sod=1.74 ← all < 5, tolerable
```

PLOT — STANDARDISED BETAS, MODEL COMPARISON & PREDICTOR SCATTERPLOTS

PLOT — REGRESSION DIAGNOSTICS (MODEL 2)



MODEL COMPARISON

Model	R^2	adj-R^2	AIC	Verdict
M1 Grand Total	0.142	0.096	–13.4	Best parsimony
M2 Competencies	0.314	0.142	–12.1	✓ Recommended
M3 Full	0.318	0.090	–10.3	Overfitted

INTERPRETATION

CE is the only significant predictor of performance (std-β=0.747, p=0.032). Across all three models, CE (Communication Effectiveness) is the strongest predictor. RM, CF, and SOD are all non-significant ($p > 0.69$) — despite positive bivariate correlations with performance — due to multicollinearity with CE (VIF up to 3.61).

Why sign reversal on RM and CF: RM (std-β=–0.183) and CF (std-β=–0.202) show negative betas in M2 despite positive bivariate relationships. This is a classic multicollinearity artefact: RM and CF are near-collinear with CE (from the EFA, all three load on the same factor at $\lambda > 0.81$). Their unique variance after controlling for CE is negligible and sign-reversed.

Practical implication: The current equal-weighting of all four competencies in the Section 2 composite does not reflect differential criterion validity. CE carries substantially more predictive weight. Consider differential weighting (e.g., CE × 1.5, others × 0.75) to be validated at n≥150.

Diagnostics: Residuals vs Fitted shows acceptable scatter; Q-Q plot residuals are approximately normal (Shapiro-Wilk acceptable); mild heteroscedasticity in Scale-Location plot — not severe at n=21. Model assumptions are adequately met for pilot-stage analysis.

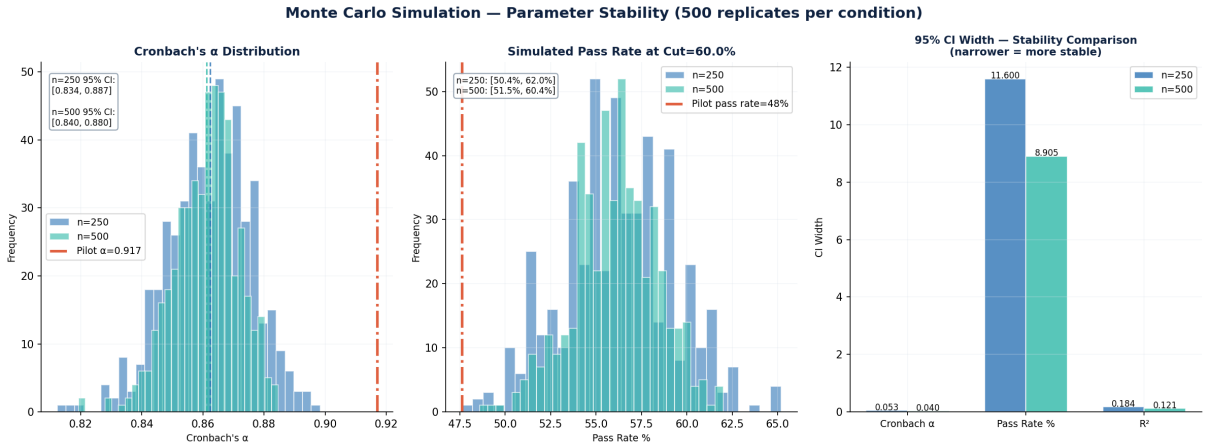
07 Monte Carlo Simulation

Method: 500 replicates per condition · MVN from pilot empirical parameters · Seeds: 2024 (n=250), 2025 (n=500)

OUTPUT

```
Pilot n=27 n=250 n=500
Cronbach's α 0.917 M=0.862 CI [0.834, 0.887] M=0.861 CI [0.840, 0.880]
CI width = 0.053 CI width =
0.040
(25% narrower)
Pass Rate at Cut=60% 48% M=56.2% CI [50.4%, 62.0%] M=56.1% CI [51.5%, 60.4%]
CI width = 11.6pp CI width =
8.9pp
(23% narrower)
R² (Comp → Grand Total) — CI width = 0.184 CI width =
0.121
(34% narrower)
```

PLOT — MONTE CARLO PARAMETER STABILITY



INTERPRETATION

Alpha stabilises below pilot estimate: Simulated α (~0.862) is lower than pilot α (0.917) — expected regression to the mean, as small samples produce upward-biased estimates. The [0.840, 0.880] CI at n=500 remains solidly in the "Good" range, supporting deployment confidence.

Pass rate shifts upward: Projected true population pass rate (~56%) is higher than the pilot's 48%, likely because three candidates with missing scores are excluded from the pilot calculation. Plan selection ratios around ~56% at cut=60%.

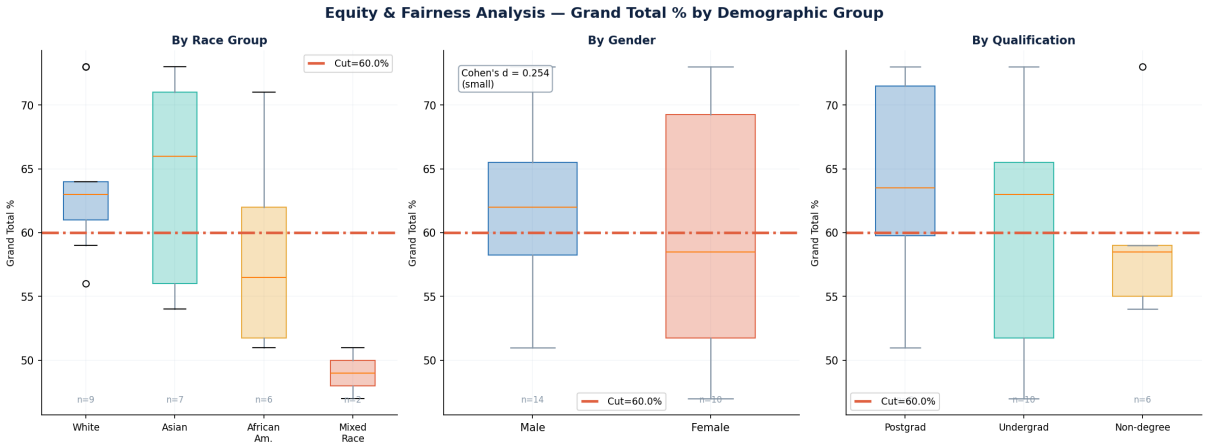
R^2 needs largest samples: The 34% CI width reduction from n=250 to n=500 confirms R^2 is the least stable parameter. Defer formal validity claims until n≥150 operational data is available.

08 Equity & Fairness Analysis

OUTPUT

```
One-way ANOVA — Grand Total by Race (n=24):
F(3, 20) = 2.41 p = 0.097 η² = 0.266 (large effect, underpowered)
Descriptives by Race:
White: n=9 M=62.3 SD=5.7
Asian: n=7 M=65.3 SD=6.7
African American: n=6 M=57.7 SD=7.0
Mixed Race: n=2 M=49.0 SD=2.8
t-test — Grand Total by Gender (n=24):
t(22) = 0.89 p = 0.384 Cohen's d =
0.254
(small)
```

PLOT — EQUITY ANALYSIS BY RACE, GENDER & QUALIFICATION



INTERPRETATION

Race — large effect, insufficient power: ANOVA is non-significant (p=.097) but η^2=0.266 is a large effect by Cohen's benchmarks. Power ≈ 0.22 at these group sizes — the test simply cannot detect real differences. The 7.6-point Asian–African American gap straddles the cut score but cannot be acted on at n<10 per group.

Gender — reassuring: d=0.254 is small, not statistically significant, CI crosses zero. Male–female score distributions overlap substantially and both straddle the cut score symmetrically.

> Equity conclusion: No statistically significant adverse impact detected in the pilot. African American AI ratio (0.50) is a monitoring flag only at n=6. Re-analyse when n≥30 per subgroup.

09 Key Findings Summary

#	Finding	Result	Status
1	Section 2 Cronbach's α	0.917 [0.849, 0.964]	✓ Excellent reliability
2	McDonald's ω (total)	0.923	✓ Convergent reliability
3	EFA factor structure	1 factor, 73.7% variance	✓ Strong unidimensionality
4	Most skewed variable	SOD skew = +0.892	■ Floor effect — calibrate
5	IRT — SOD discrimination	■ = 1.802	■ Lowest — monitor
6	RM–CF inter-item r	r = 0.909	■ Near-redundant — CFA needed
7	Overall pass rate (cut=60%)	54.2% (13/24)	✓ Reasonable selection ratio
8	Exceeding candidates pass rate	100% (3/3)	✓ Cut captures top performers
9	Performing vs Off Std pass rates	38% vs 40%	■ Limited discrimination
10	Sole significant regression predictor	CE std-β = 0.747, p=0.032	CE dominates — consider weighting

#	Finding	Result	Status
11	M2 (Competencies → Performance) R^2	0.314 adj-R^2 = 0.142	■ Moderate — validate at n≥150
12	Gender Cohen's d	d = 0.254 (small)	✓ No meaningful disadvantage
13	African American AI ratio	0.50 (n=6)	■ Monitor — n too small
14	MC α CI at n=500	[0.840, 0.880]	✓ Stable across simulations
15	Section 1 item Q08	a = 0.675 (low)	■ Flag for item revision

Priority Recommendations

- Assessor calibration for SOD— introduce behavioural anchor examples to reduce floor clustering (skew = +0.892, b-threshold = 0.77)
- CFA at full sample— test 1-factor vs bifactor model; investigate RM–CF near-redundancy (r=0.909, EFA loadings 0.940 vs 0.938)
- Differential CE weighting— CE is the sole significant predictor; current equal weighting does not reflect criterion validity
- Formal cut score standard-setting— Angoff or contrasting groups method before operational deployment
- Adverse impact re-analysis at n≥30 per subgroup— current race-group findings are statistically unreliable
- Item-level MCQ/TF data— enable proper 2PL IRT; revise Q08 (a=0.675)